Windows Computer Literacy
Setup and Everyday Reference

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1 Install VS Code or VSCodium

Choose one editor. The VS Code User Setup or VSCodium official installation options; Or use WinGet to install either editor. Winget install vscode –source winget. Winget install vscodium –source winget.

2 Install the shared tools

- Install Python following the instructions at official Windows downloads or with PowerShell using "winget install Python.Python.3.14".
- Install Git following the instructions at the official Git downloads or with PowerShell using "winget install –id Git.Git -e –source winget".
- I have found many people have trouble with LaTeX in vscode on windows, i would still recomend this as it keeps all your things in one place. you can try MiKTeX or TeX Live. If none of this works and you dont care to install a local TeX distribution, use Overleaf is a browser alternative, this will cause problems down the line when you have longer document and it begins to ask for money. Other dedicated programs are available as well but i have no experience with them.

After installation you should verify that the tools are correctly installed by checking their versions. Close and reopen PowerShell after installation and type the following commands to check the versions of the installed tools.

```
py --version
git --version
Get-Location
```

If `py` is unavailable but `python -version` identifies the installed interpreter, use `python` in the environment commands below. You may have to write python3 instead of python depending on the version

3 PowerShell, projects, and paths

This is for PowerShell (Windows Terminal), not Command Prompt, Git Bash, or WSL, for these instructions. A terminal runs a shell to navigate folders and start programs. Relative paths begin from the current working directory. In File Explorer, enable filename extensions so a source file does not accidentally become `report.tex.txt`. Keep projects in a writable user folder.

To install WSL use the command `wsl -install` in the powershell

```
pendulum-lab/
  README.md
  requirements.txt
  data/
    raw_measurements.csv
  analysis/
    analysis.py
    analysis.ipynb
  figures/
  report/
    report.tex
```

Task	PowerShell command
Current directory	`Get-Location` (alias: `pwd`)
List files, including hidden	`Get-ChildItem -Force`
Move to the different directory	`Set-Location "DirectoryFilePath"`
Enter a path with spaces	`Set-Location "My Project"`
Copy a file	`Copy-Item source.csv copy.csv`
Move a file	`Move-Item copy.csv data.csv`
Find CSV files	`Get-ChildItem -Filter *.csv -Recurse`
Search file text	`Select-String -Path notes.txt -Pattern time`
Open the current folder	`Start-Process .`
Open the folder in the editor	`"code ."` or `"codium ."`

Inspect paths before copying or moving files: an existing destination may be overwritten. Use File Explorer's Recycle Bin when removing files. If the editor command is unavailable, use File > Open Folder. For more on PowerShell's file and directory commands, see Microsoft's working with files and folders guide.

4 Python and project environments

From the project directory, create a virtual environment and install packages using its interpreter:

```
py -m venv .venv
.\.venv\Scripts\python.exe -m pip install numpy pandas matplotlib etc
.\.venv\Scripts\python.exe -m pip check
.\.venv\Scripts\python.exe -c "import sys; print(sys.executable)"
```

The printed path should include `.venv`. Optional PowerShell activation is:

```
.\.venv\Scripts\Activate.ps1
```

After activation, use `python -m pip ...`; run `deactivate` to leave the environment. If activation is blocked, call the environment's Python executable directly rather than weakening the execution policy. A virtual environment keeps project packages separate; it is not a security sandbox, and it should not be committed or copied to another computer. See the Python virtual-environment documentation.

5 Git and LaTeX

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I just found this Complete Guide Ingore my guide, im just to lazzy to edit this document properly

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Git works locally; GitHub hosts online copies and collaboration. Saving a file is not the same as committing it, and a commit is not the same as pushing it. For the shared commands and safe workflow, see the Git and GitHub guide.

LaTeX source (`.tex`) is compiled into a PDF. Use Overleaf in a browser or install MiKTeX/TeX Live for local builds. From the repository root, build this handout with:

```
latexmk -cd -pdf -interaction=nonstopmode -halt-on-error Windows/Windows.TEX
```

The shared LaTeX guide covers report structure, equations, figures, and references; see also the Overleaf LaTeX introduction and LaTeX Project documentation.

6 Shortcuts and PowerShell profile

- `Ctrl+C` interrupts a command; `Tab` completes paths; `Up` recalls command history.
- Windows Terminal commonly uses `Ctrl+Shift+C/V` for copy/paste.
- `Win+E` opens File Explorer; `Win+Left/Right` tiles a window to the side.

PowerShell reads a profile script when it starts. Display its location with `$PROFILE`. Create the parent folder and file only if needed, then open the profile in Notepad:

```
New-Item -ItemType Directory -Path (Split-Path -Parent $PROFILE) -Force | Out-Null
if (!(Test-Path -Path $PROFILE)) {
  New-Item -ItemType File -Path $PROFILE | Out-Null
}
notepad $PROFILE
```

Add personal shortcuts, save, and load them into the current session with

```
function ll { Get-ChildItem -Force }
function croot { Set-Location $HOME }
. $PROFILE
```

Back up and review your profile before editing it. Bash aliases from `.bashrc` do not automatically work in PowerShell. See Microsoft's PowerShell profile documentation.
For Python, fitting, and data-analysis exercises, use the shared exercise notebook.